

```

> # chapter3_homework.R
> # R语言第三章课后作业：基本数据管理
> # 题目0：创建数据框
> manager <- c(1, 2, 3, 4, 5)
> date <- c("10/24/08", "10/28/08", "10/01/08", "10/12/08", "05/01/09")
> country <- c("US", "US", "UK", "UK", "UK")
> gender <- c("M", "F", "F", "M", "F")
> age <- c(32, 45, 25, 39, 99)
> q1 <- c(5, 3, 3, 3, 2)
> q2 <- c(4, 5, 5, 3, 2)
> q3 <- c(5, 2, 5, 4, 1)
> q4 <- c(5, 5, 5, NA, 2)
> q5 <- c(5, 5, 2, NA, 1)
> leadership <- data.frame(manager, date, country, gender, age,
+                             q1, q2, q3, q4, q5)

```

```

> # 题目1

```

```

> head(leadership)

```

	manager	date	country	gender	age	q1	q2	q3	q4	q5
1	1	10/24/08	US	M	32	5	4	5	5	5
2	2	10/28/08	US	F	45	3	5	2	5	5
3	3	10/01/08	UK	F	25	3	5	5	5	2
4	4	10/12/08	UK	M	39	3	3	4	NA	NA
5	5	05/01/09	UK	F	99	2	2	1	2	1

```

> str(leadership)

```

```

'data.frame':   5 obs. of  10 variables:
 $ manager: num  1 2 3 4 5
 $ date   : chr  "10/24/08" "10/28/08" "10/01/08" "10/12/08" ...
 $ country: chr  "US" "US" "UK" "UK" ...
 $ gender : chr  "M" "F" "F" "M" ...
 $ age    : num  32 45 25 39 99
 $ q1     : num  5 3 3 3 2

```

```

$ q4      : num  5 5 5 NA 2
$ q5      : num  5 5 2 NA 1
> names(leadership)
[1] "manager" "date"      "country" "gender"  "age"      "q1"       "q2"       "q3"       "q4"
[10] "q5"
> # 题目2
> leadership$total_score <- leadership$q1 + leadership$q2 + leadership$q3 +
+ leadership$q4 + leadership$q5
> # 题目3
> leadership$mean_score <- rowMeans(leadership[, c("q1", "q2", "q3", "q4", "q5")],
+                                   na.rm = TRUE)
> # 题目4
> leadership$age[leadership$age == 99] <- NA
> # 题目5
> leadership$agecat <- NA
> leadership$agecat[leadership$age < 35] <- "Young"
> leadership$agecat[leadership$age >= 35 & leadership$age <= 50] <- "Middle"
> leadership$agecat[leadership$age > 50] <- "Senior"
> # 题目6
> leadership$date <- as.Date(leadership$date, format = "%m/%d/%y")
> # 题目7
> leadership$days_to_2009 <- as.numeric(leadership$date - as.Date("2009-01-01"))
> # 题目8
> leadership$gender <- factor(leadership$gender,
+                             levels = c("M", "F"),
+                             labels = c("Male", "Female"))
> # 题目9
> colSums(is.na(leadership))

```



```
> colSums(is.na(leadership))
```

manager	date	country	gender	age	q1	q2
0	0	0	0	1	0	0
q3	q4	q5	total_score	mean_score	agecat	days_to_2009
0	1	1	1	0	1	0

```
> # 题目10
> leadership_complete <- na.omit(leadership)
> # 题目11
> leadership_age_order <- leadership[order(leadership$age), ]
> # 题目12
> leadership_score_order <- leadership[order(-leadership$mean_score), ]
> # 题目13
> leadership_under40 <- leadership[leadership$age < 40, ]
> # 题目14
> leadership_female_high <- leadership[leadership$gender == "Female" &
+                                     leadership$mean_score > 4, ]
> # 题目15
> leadership_sub <- leadership[, c("manager", "country", "gender",
+                                   "age", "total_score", "mean_score")]
> # 题目16
> names(leadership)[names(leadership) == "manager"] <- "managerID"
> # 题目17
> library(dplyr)
> leadership_dplyr1 <- leadership %>%
+   mutate(
+     age = ifelse(age == 99, NA, age),
+     mean_score = rowMeans(select(., q1, q2, q3, q4, q5), na.rm = TRUE)
+   ) %>%
+   filter(!is.na(age)) %>%
+   select(manager, country, gender, age, mean_score)
```

Warning message:

Using an external vector in selections was deprecated in tidyselect 1.1.0.

i Please use ``all_of()`` or ``any_of()`` instead.

Was:

```
data %>% select(manager)
```

Now:

```
data %>% select(all_of(manager))
```

See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.

This warning is displayed once per session.

Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.

```
> leadership_dplyr1
```

	managerID	date	country	gender	age	mean_score
1	1	2008-10-24	US	Male	32	4.800000
2	2	2008-10-28	US	Female	45	4.000000
3	3	2008-10-01	UK	Female	25	4.000000
4	4	2008-10-12	UK	Male	39	3.333333

```
> # 题目18
```

```
> leadership_dplyr2 <- leadership %>%
```

```
+ mutate(
```

```
+   total_score = q1 + q2 + q3 + q4 + q5,
```

```
+   mean_score = rowMeans(select(., q1, q2, q3, q4, q5), na.rm = TRUE)
```

```
+ ) %>%
```

```
+ filter(mean_score >= 4) %>%
```

```
+ select(manager, gender, age, total_score, mean_score) %>%
```

```
+ arrange(desc(mean_score))
```

```
> leadership_dplyr2
```



```

> # 题目18
> leadership_dplyr2 <- leadership %>%
+   mutate(
+     total_score = q1 + q2 + q3 + q4 + q5,
+     mean_score = rowMeans(select(., q1, q2, q3, q4, q5), na.rm = TRUE)
+   ) %>%
+   filter(mean_score >= 4) %>%
+   select(manager, gender, age, total_score, mean_score) %>%
+   arrange(desc(mean_score))
> leadership_dplyr2
  managerID      date country gender age total_score mean_score
1          1 2008-10-24      US  Male  32           24         4.8
2          2 2008-10-28      US Female  45           20         4.0
3          3 2008-10-01      UK Female  25           20         4.0
> mean(leadership$q4, na.rm = TRUE)
[1] 4.25
>
> # 题目19
> # 题目19
> # 这段代码的含义是：计算 leadership 数据框中 q4 这一列的平均值。
> # 题目19
> # 这段代码的含义是：计算 leadership 数据框中 q4 这一列的平均值。
> # 其中 na.rm = TRUE 的作用是“在计算平均值时删除缺失值 NA”。
> # 题目19
> # 这段代码的含义是：计算 leadership 数据框中 q4 这一列的平均值。
> # 其中 na.rm = TRUE 的作用是“在计算平均值时删除缺失值 NA”。
> # 如果不写 na.rm = TRUE，只要 q4 中存在 NA，计算结果就会返回 NA。
> leadership$age >= 35 && leadership$age <= 50

```

```
> # 这段代码的含义是：计算 leadership 数据框中 q4 这一列的平均值。  
> # 其中 na.rm = TRUE 的作用是“在计算平均值时删除缺失值 NA”。  
> # 如果不写 na.rm = TRUE，只要 q4 中存在 NA，计算结果就会返回 NA。  
> leadership$age >= 35 && leadership$age <= 50
```

```
Error in leadership$age >= 35 && leadership$age <= 50 :  
  'length = 5' in coercion to 'logical(1)'
```

 Show Traceback
 Rerun with Debug

```
> leadership$age >= 35 & leadership$age <= 50  
[1] FALSE TRUE FALSE TRUE NA  
> leadership$age >= 35 & leadership$age <= 50  
[1] FALSE TRUE FALSE TRUE NA  
>  
> # 题目21  
> write.csv(leadership_dplyr2, "chapter3_result.csv", row.names = FALSE)  
> |
```